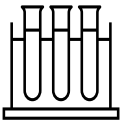


Domain Term of the Week

Episode 4 : Ore Grade Optimization

Ore Grade Optimization refers to the process of maximizing the economic value of mined material by efficiently identifying, classifying, and processing ore based on its metal content (grade). It ensures that higher-value material is prioritized while minimizing waste and dilution.

Activities involved in the term



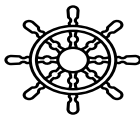
Ore Sampling & Analysis

Collecting and testing ore samples to determine mineral grades.



Geological Modeling

Creating geological models to identify high-value ore zones.



Grade Control

Monitoring ore quality during extraction and processing.



Ore Blending

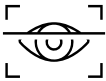
Combining ores of different grades to achieve optimal feed quality.



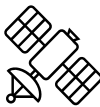
Processing Optimization

Adjusting plant operations based on ore characteristics.

Support Activities



Real-time monitoring: Tracking ore quality and production metrics through dashboards.



Geospatial & Geological Systems
Supporting mine planning using GIS and geological data.

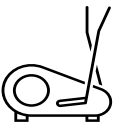


Advanced Analytics :
Using AI/ML models to predict ore quality and recovery rates.



Reporting & Performance Analytics :
Generating production, recovery, and profitability insights.

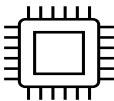
Benefits to the Organization



Higher Copper Recovery



Maximizes extraction of valuable minerals.



Improved Processing Efficiency



Ensures optimal use of available ore reserves.



Reduced Operational Costs



Better Resource Utilization